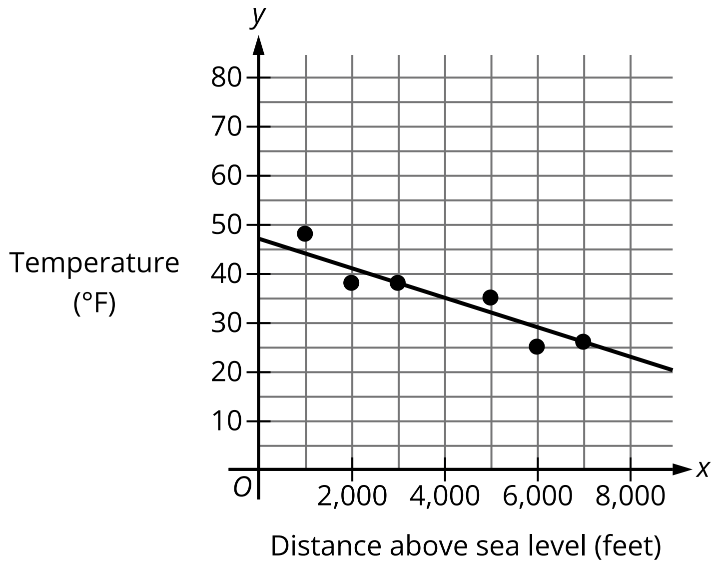
Question ID: bc59c2d9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Easy

Question

The scatterplot shows the temperature, in degrees Fahrenheit $(°F)$, and the distance above sea level, in feet, measured at 6 locations on Mount Jefferson. A line of best fit is also shown.



At a distance of $4{,}000$ feet above sea level, what is the temperature, in $°F$, predicted by the line of best fit?

Answer

A. 47

B. 35

C. 25

D. 0

Correct Answer: B

Rationale

Choice B is correct. In the given scatterplot, the *x*-values represent the distance above sea level, in feet, and the *y*-values represent the temperature, in $°F$. The point on the line of best fit with an *x*-value of $4{,}000$ has a corresponding *y*-value of 35. Therefore, at a distance of $4{,}000$ feet above sea level, the temperature predicted by the line of best fit is $35°F$.

Choice A is incorrect. This is the temperature, in $°F$, predicted by the line of best fit at a distance of 0 feet above sea level.

Choice C is incorrect. This is the measured temperature, in $°F$, at a distance of $6{,}000$ feet above sea level.

Choice D is incorrect and may result from conceptual or calculation errors.